

Safety

The rules that do not bend

Read this completely before you build. Post it where you fire.

WHAT YOU ARE BUILDING

A potato leaving a combustion launcher carries energy in the same range as a handgun bullet — less speed, far more mass. It will travel several hundred feet, it will go through a window, a fence panel or a windscreen, and it will kill a person or an animal it hits. The chamber holds a pressure spike of several atmospheres for a few thousandths of a second. Treat both ends of this object with the seriousness they have earned.

The six rules

- 1 One adult owns the fuel and the igniter.** Named in advance, and it does not rotate. Nobody else carries the fuel can, sprays the chamber, or presses the igniter. Ever.
- 2 Muzzle discipline, exactly as for a rifle.** The muzzle points downrange or at the ground, at all times, loaded or not, fuelled or not. Never swept across a person. Never rested pointing at anything you would mind destroying. There is no such thing as an unloaded launcher.
- 3 Confirm the range is clear before every shot** — and mean the whole flight path, not just the first fifty yards. A potato at 45 degrees can carry several hundred feet and it does not stop at your property line.
- 4 Everyone behind the firing line.** One line, marked on the ground, behind the breech. The person firing is the only one forward of it, and only when everyone else has said they are clear.
- 5 Never look down the barrel, and never stand beside the chamber.** If it fails, the chamber is what fails, and it fails sideways. The person firing stands behind the breech end, not level with it.
- 6 A misfire is not over.** If it does not fire, wait a full **sixty seconds** with the muzzle downrange before anyone touches anything. Then open the cleanout to vent, and only then approach the muzzle. Delayed ignition is real and it is how people get hurt.

Build it safe, not just operate it safe

Pressure-rated pipe only	Schedule 40 pressure PVC, marked with an NSF-PW rating and a pressure number. DWV, foam-core and cellular-core pipe are not rated for internal pressure and fail by shattering. This is the single most important line on this sheet.
Cure it fully	Solvent-welded joints reach full strength in 48 hours , not in the twenty minutes it takes them to stop feeling wet. A joint fired at four hours is a joint that can let go. Write the date on it.
Never compressed air	This design is for combustion only. PVC under sustained pneumatic pressure stores far more energy and fails catastrophically, and the failure mode is fragmentation. Do not connect a compressor to this, ever, no matter what you read.
Keep it warm	PVC becomes brittle in the cold. Do not fire below about 40 °F.
Inspect before every session	Look for crazing, whitening at the joints, star cracks, or any deformation of the chamber. Any of them retires the launcher permanently.
Retire it on schedule	PVC fatigues. This is a consumable object, not an heirloom. Rebuild the chamber periodically rather than finding out where its limit is.

The fuel is the part people get wrong

Less than you think	Over-fuelling is the most common cause of a burst chamber and of a launcher that will not fire. A rich mixture does not burn hotter — it does not burn at all, because there is not enough oxygen. <i>see the 2. Combustion sheet</i>
A measured count	Decide a spray count — a one-second burst is a reasonable starting point for a 3 in chamber — write it on the launcher, and keep to it. Do not adjust it by feel between shots.
Never top up a failed shot	If it does not fire, the chamber is already full of unburnt fuel. Adding more guarantees a rich mixture that still will not fire, and eventually a chamber full of fuel that fires all at once. Vent it, fan it out, start again.
Never a liquid accelerant	No starter fluid, no petrol, no propane torch bottles. Aerosol hairspray is chosen precisely because it is hard to deliver a genuinely dangerous quantity in one burst.
Residue builds up	Hairspray leaves a sticky film that collects and eventually interferes with ignition. Wipe the chamber out through the cleanout regularly.
Store fuel away from the launcher	And well away from the firing line and any source of ignition.

Choosing the ground

Clear distance	Several hundred yards of open ground downrange, with a wide margin either side. Assume the thing will fly twice as far as you expect, because one day it will.
A backstop	An earth bank, a hillside, a treeline with nothing behind it. Something that stops a potato that goes further than planned.
Nothing downrange	No roads, dwellings, outbuildings, livestock, footpaths or property lines in the cone of fire. Walk it before the first shot of the day.
Know where it lands	Have a spotter, and do not fire the next one until the last one has been accounted for.
Fire risk	An ignition source and dry grass. Do not fire in drought conditions, and keep water or an extinguisher on the line.
Check the law	These are legal in most of the United States and restricted in some places, and local ordinance may differ from state law. Confirm before you build, not after somebody complains.

Before every shot — read it aloud

1. Is the range walked and clear, all the way to the backstop?
2. Is everybody behind the firing line and do they all know a shot is coming?
3. Eyes and ears on, everyone?
4. Is the chamber inspected, and is the cleanout screwed fully home?
5. Is the muzzle pointing downrange, and has it been the whole time?
6. Is this the measured fuel dose, and is this the first dose for this shot?
7. Is the last projectile accounted for?
8. Am I the named adult, and am I the one about to press the igniter?

IF THE CHAMBER EVER FAILS

Stop for the day. Do not repair it, do not glue it, do not tape it. A PVC chamber that has split or crazed once has told you its history of stress, and the next failure will be worse. Cut it up so it cannot be reused, and build a new one — which is cheap, and is the reason the design is made of plumbing parts in the first place.

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